

Azure Cosmos DB Change Feed Trigger Using Azure Functions

Summary:-

In this lab, you will enable the Azure Cosmos DB Change Feed feature and create an Azure Functions project that processes document changes using the Azure Cosmos DB Trigger. Instead of repeatedly querying the database for updates, Azure Functions automatically responds whenever new documents are created or existing documents are modified, enabling efficient event-driven processing.

Let's Start:-

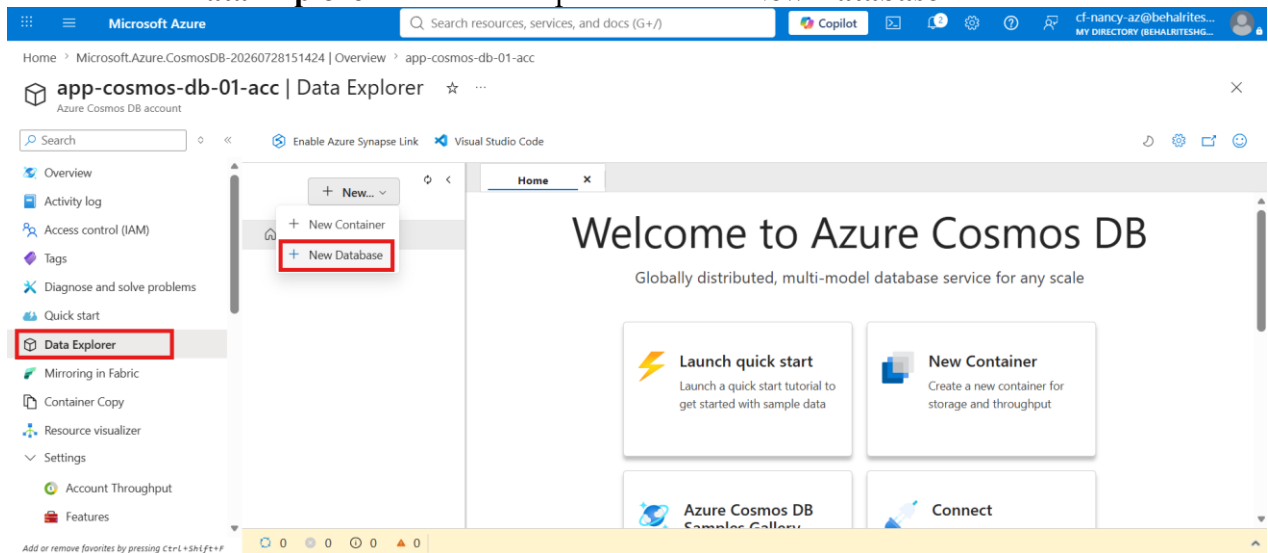
After completing this lab, you will be able to:

- Enable the Azure Cosmos DB Change Feed feature.
- Create an Azure Functions project in Visual Studio Code.
- Configure an Azure Cosmos DB Trigger.
- Connect the trigger to an Azure Cosmos DB container.
- Understand how Azure Functions process Change Feed events automatically.

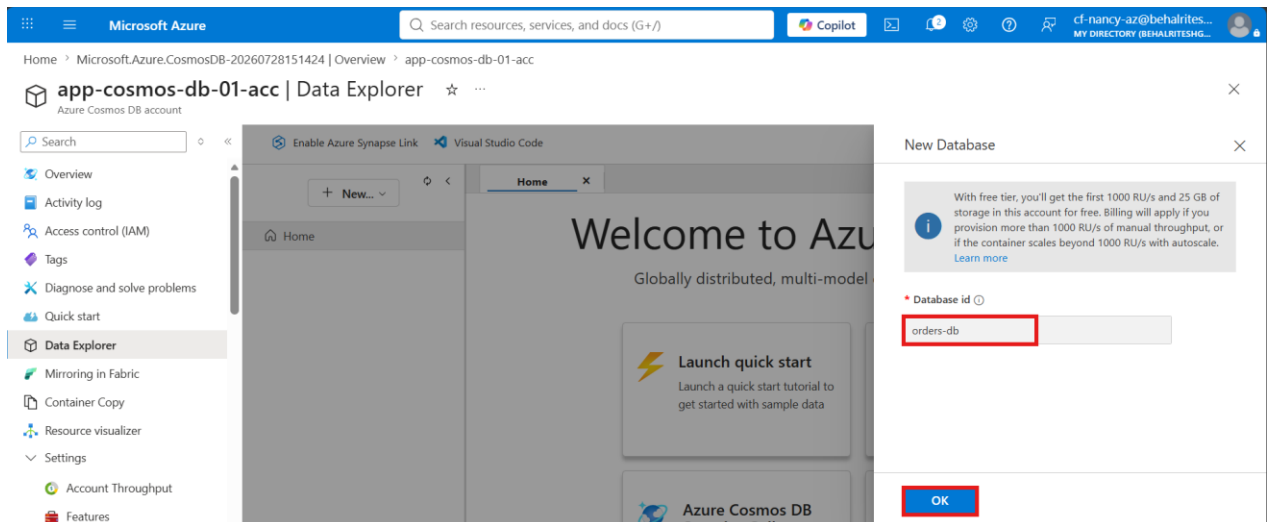
Lab Steps:-

Step 1 – Enable the Azure Cosmos DB Change Feed Feature

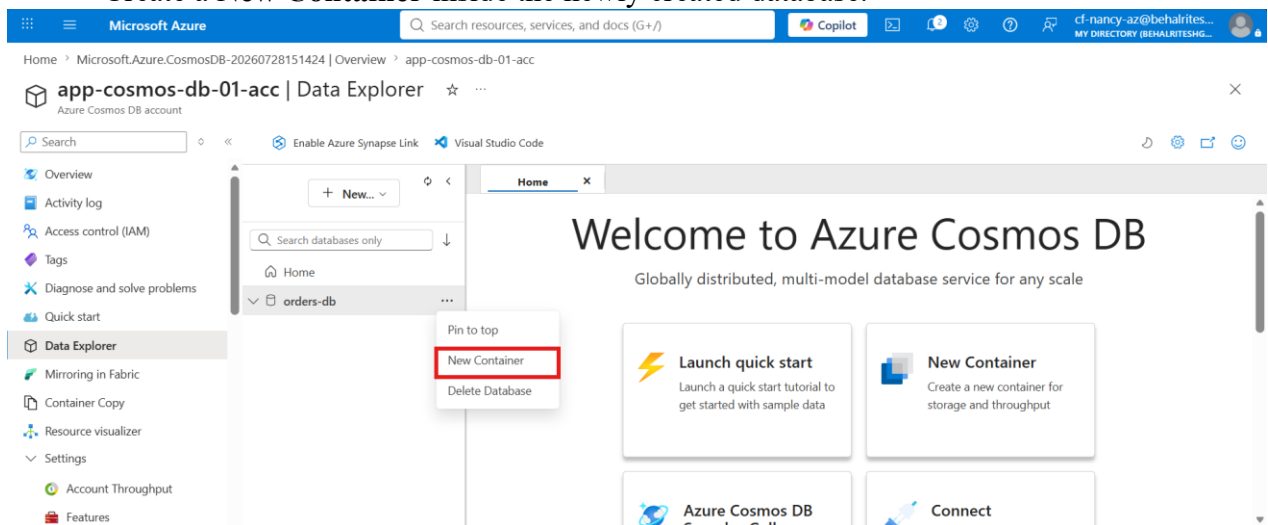
- Sign in to the Azure portal.
- Open your Azure Cosmos DB for NoSQL account.
- Go to **Data Explorer** from the left pane. Create a **New Database**.



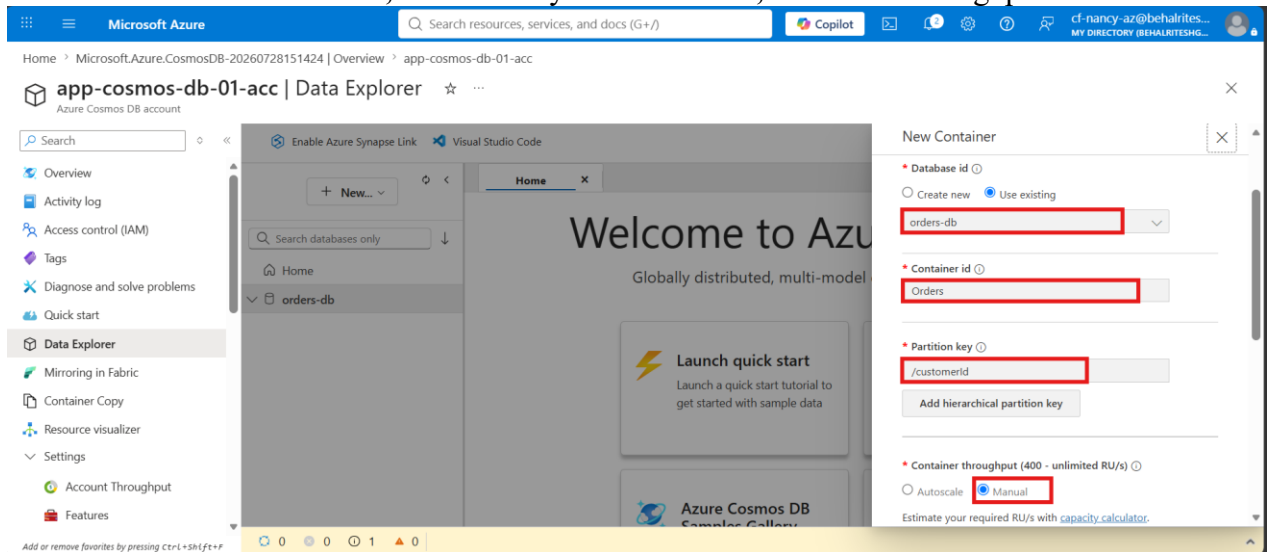
- Name it **orders-db**.



- Create a New Container inside the newly created database.



- Enter:
Container id: **Orders** , Partition key: **/customerId** , Container throughput: **Manual**.



- From the left pane, Expand **Settings**. Select **Backup & Restore**.
- Select **Change to continuous mode**. Choose **Continuous (7 days)** and save.

Microsoft Azure

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app-cosmos-db-01-acc | Backup & Restore

Azure Cosmos DB account

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Replicate data globally

Default consistency

Backup & Restore

Networking

Add or remove favorites by pressing Ctrl+Shift+F

Your account is on periodic backup mode. You can now change to continuous mode for a better experience. [Change to continuous mode](#)

Backup policy mode (change)

Periodic

Backup Interval

How often would you like your backups to be performed?

240 Minute(s)

60-1440

Backup Retention

How long would you like your backups to be saved?

8 Hours(s)

8-720

Copies of data retained 2

Backup Storage Redundancy

Submit Discard

Backup policy mode

Azure Cosmos DB provides three different backup policies. You will not be able to switch to Periodic mode once you adopt Continuous mode. [Learn more](#) about the differences of the backup policies and pricing details.

☐ Periodic

Backup is taken at periodic interval based on your configuration

☒ Continuous (7 days)

Provides backup window of 7 days / 168 hours and you can restore to any point of time within the window. This mode is available for free.

☐ Continuous (30 days)

Provides backup window of 30 days / 720 hours and you can restore to any point of time within the window. This mode has cost impact.

☐ Continuous (35 days)

Provides backup window of 35 days / 840 hours and you can restore to any point of time within the window. This mode has cost impact.

Once you change to Continuous mode, you will not be able to change the backup policy to Periodic mode.

Save Cancel

- From the left pane, select **Features**.
- Enable the **Change Feed** feature (if it is disabled).

Microsoft Azure

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app-cosmos-db-01-acc | Features

Azure Cosmos DB account

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Point In Time Restore

Vaulted Backup

Add or remove favorites by pressing Ctrl+Shift+F

Refresh Feedback

Feature

All versions and deletes change feed mode

Burst Capacity

Diagnostics full-text query

Dynamic Data Masking

Dynamic Scaling (Per Region and Per Partition Autoscale)

Global Secondary Index for NoSQL API

New Features for Full Text Search

Partition merge (preview)

Per-Partition Automatic Failover

Priority based execution

Vector Search for NoSQL API

The Azure Cosmos DB all versions and deletes change feed mode provides a record of changes to items from create, update, and delete operations. You will get a record of each change to items in the order that it occurred, including intermediate changes to an item between change feed reads.

To read from the change feed in all versions and deletes mode, you must have continuous backups configured for your Azure Cosmos DB account. In this mode, you will only be able to read changes that occurred within the continuous backup period configured for the account.

Please note that enabling all versions and deletes change feed mode can take up to 30 minutes. After the feature is enabled, you can begin reading changes in all versions and deletes change feed mode. Enabling this mode has no impact on existing change feed workloads, and latest version change feed mode will continue to function.

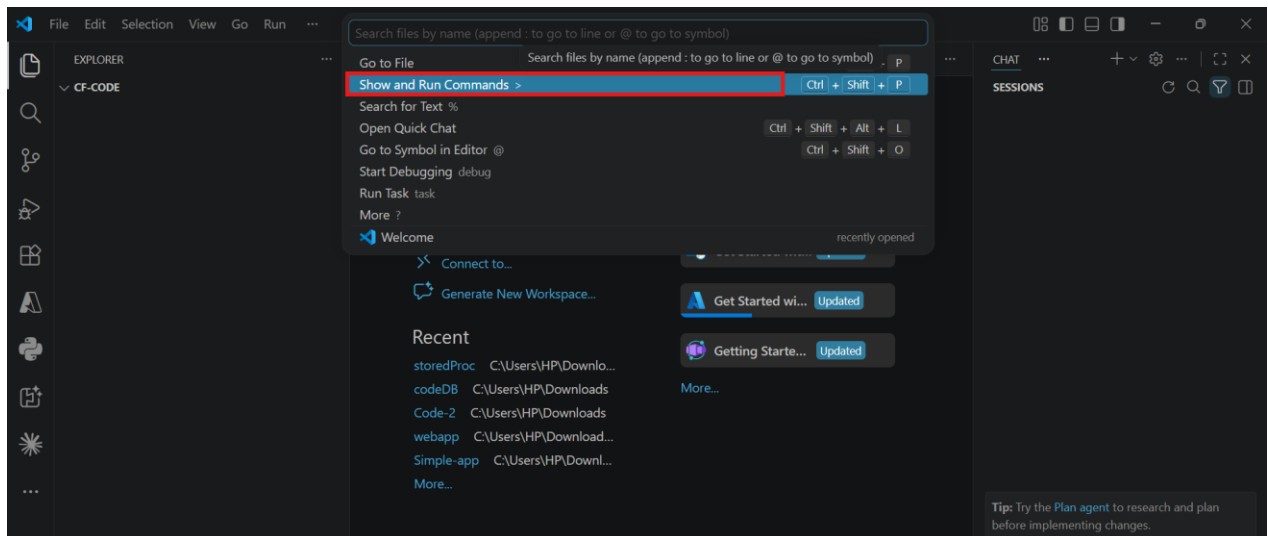
[Learn More](#)

Enable Cancel

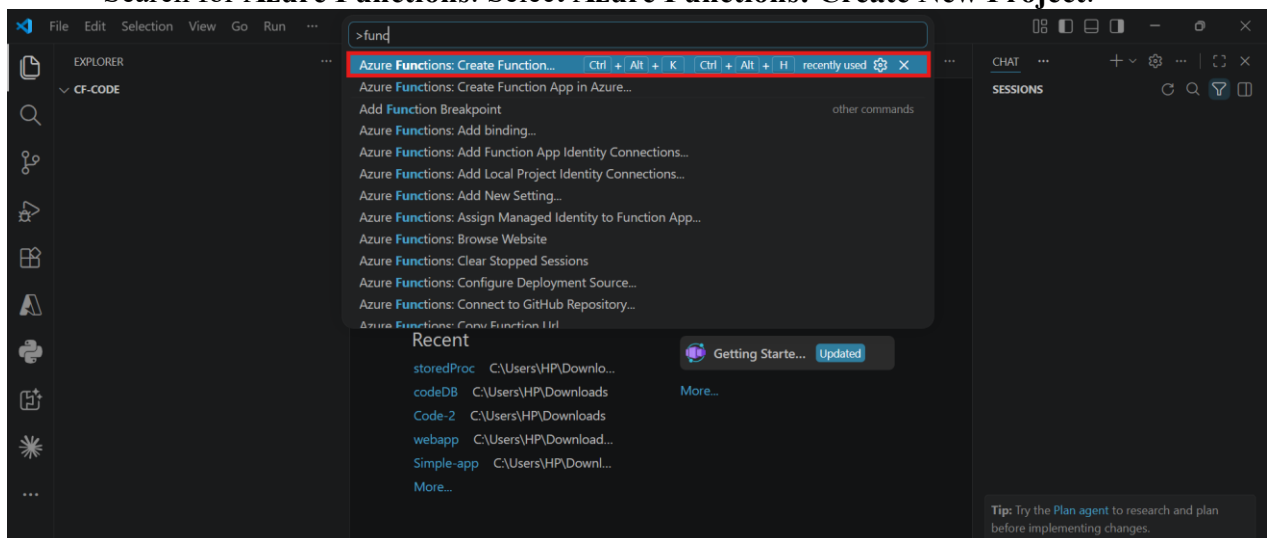
- Save the configuration.
- The Change Feed continuously records document inserts and updates, allowing applications to react automatically whenever data changes.

Step 2 – Create an Azure Functions Project

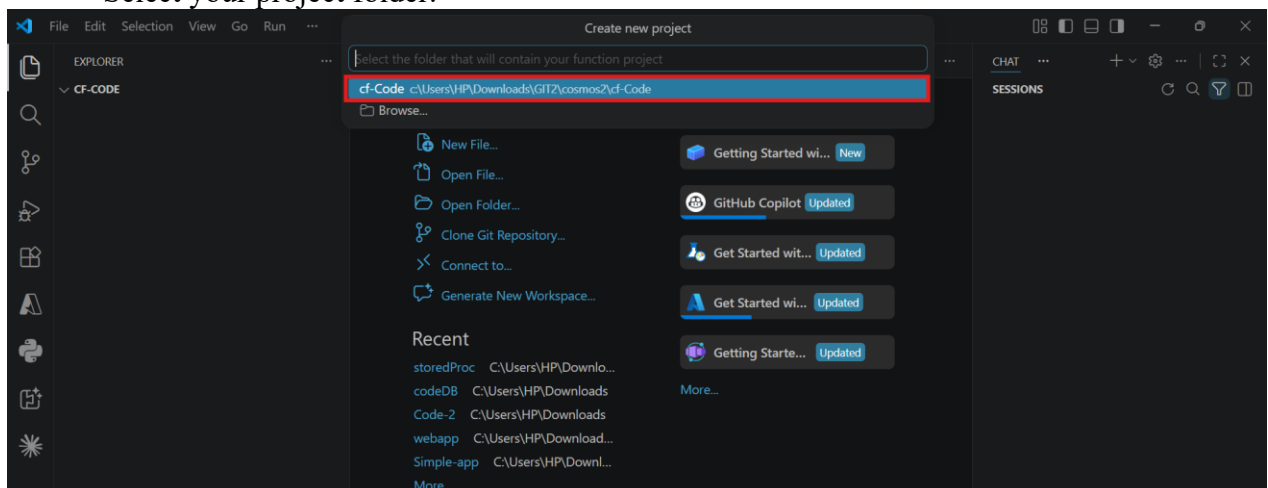
- Open Visual Studio Code.
- Open an empty project folder.
- Press **Ctrl + Shift + P** or select **Show and Run Commands >** in the Command Palette.



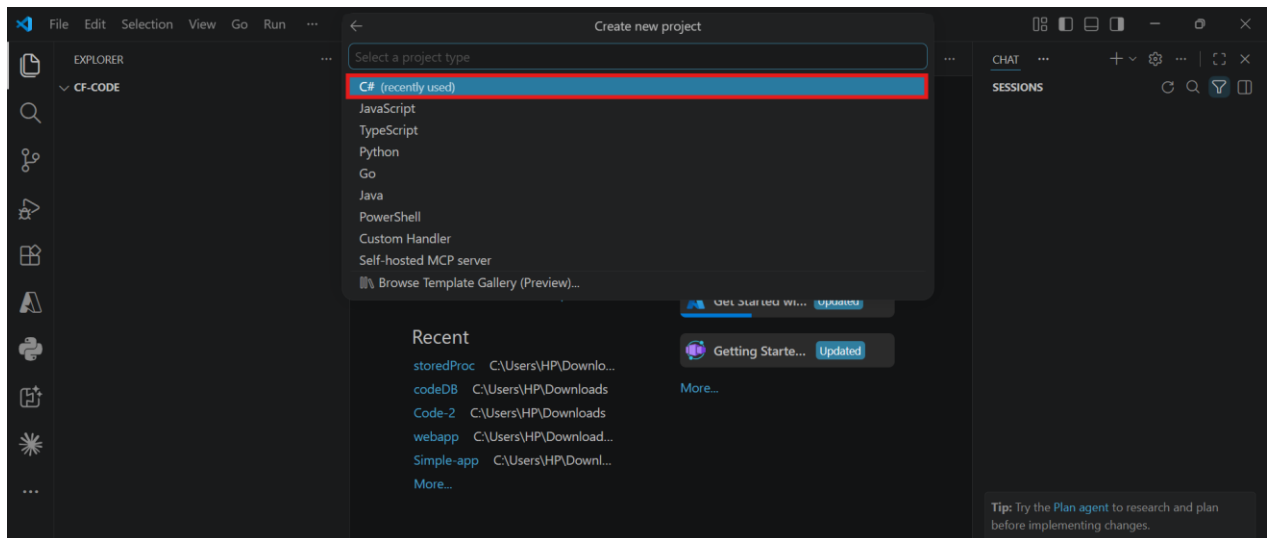
- Search for Azure Functions. Select **Azure Functions: Create New Project**.



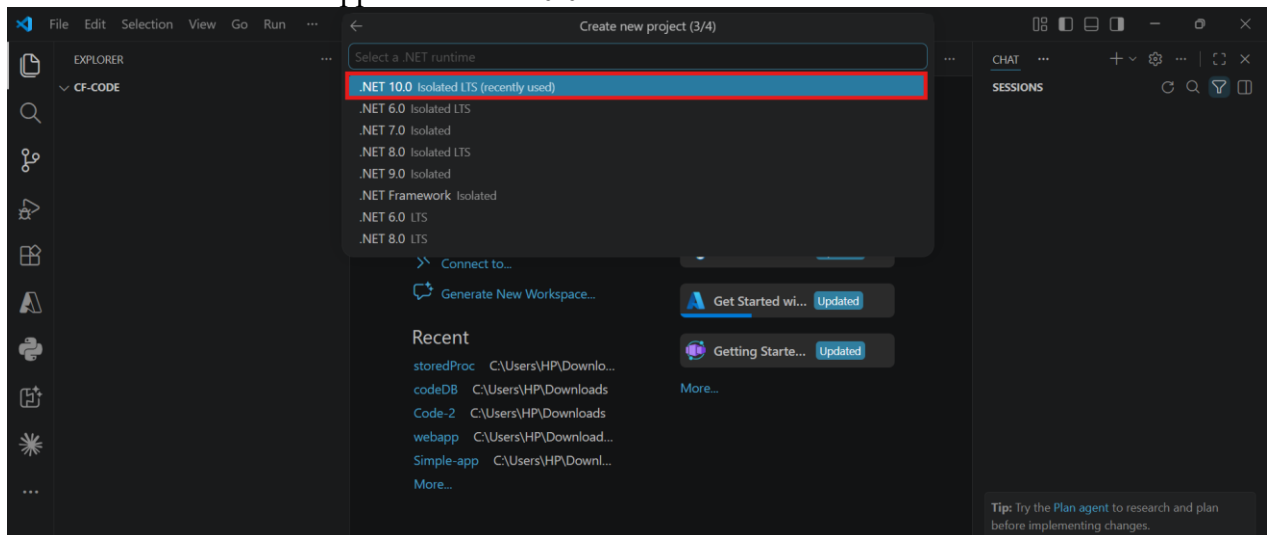
- Select your project folder.



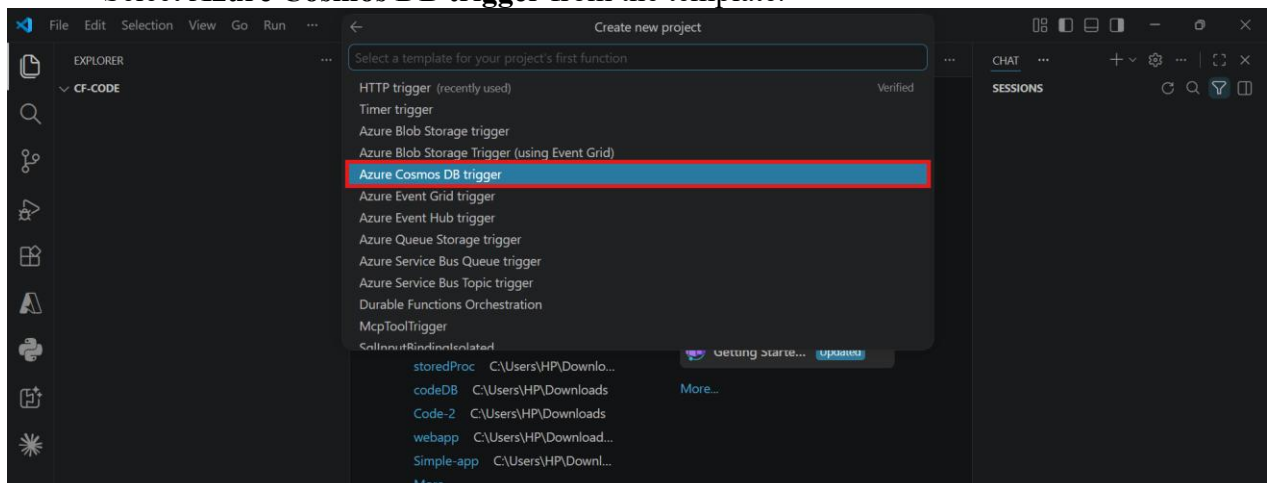
- Choose C# as the language.



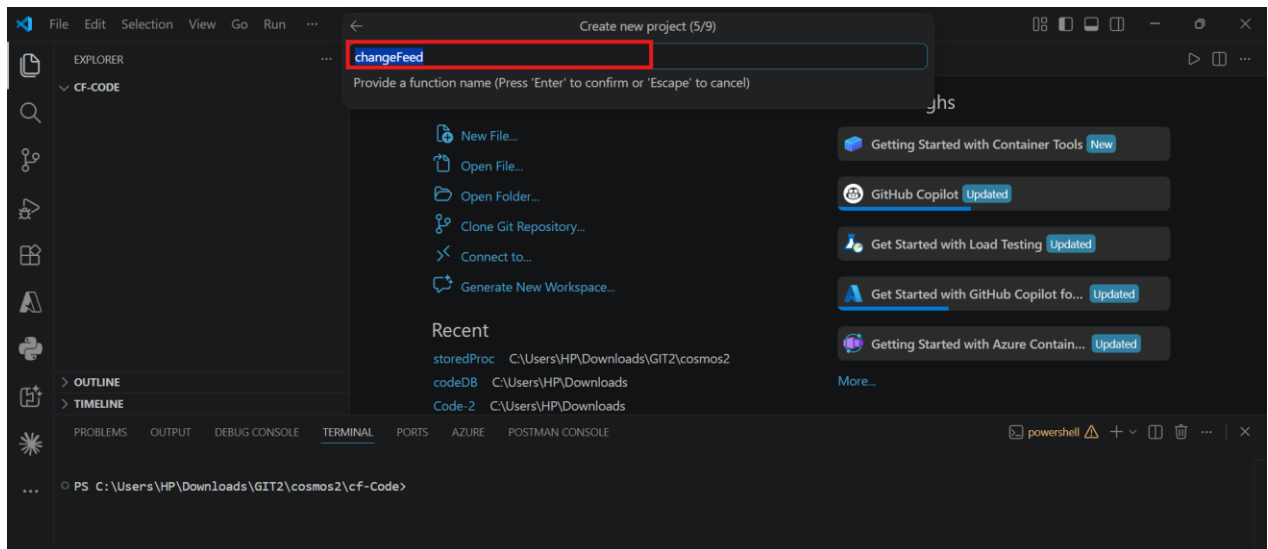
- Select the latest supported .NET 10.0 version.



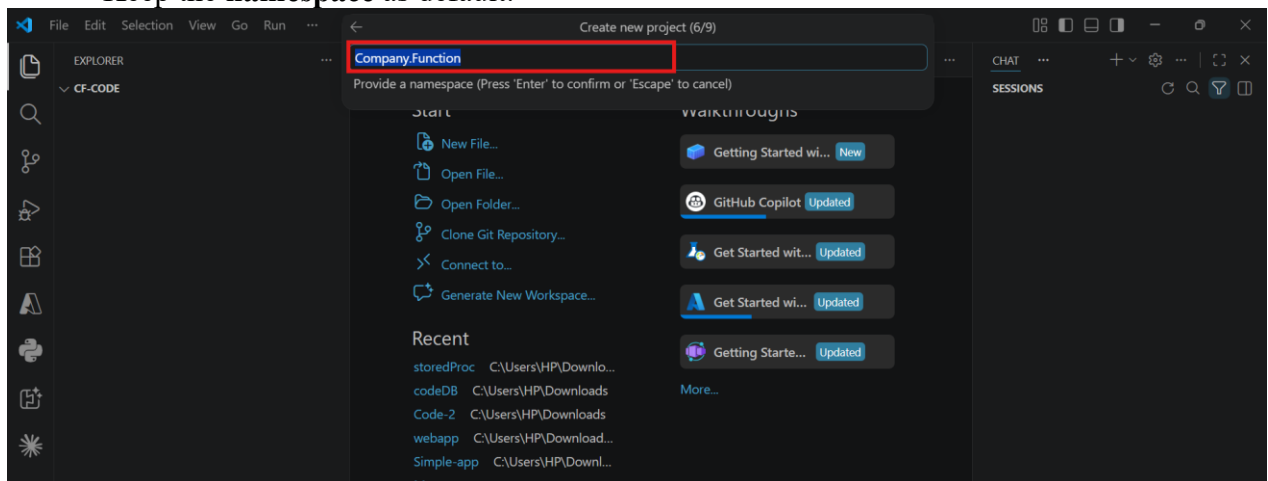
- Select Azure Cosmos DB trigger from the template.



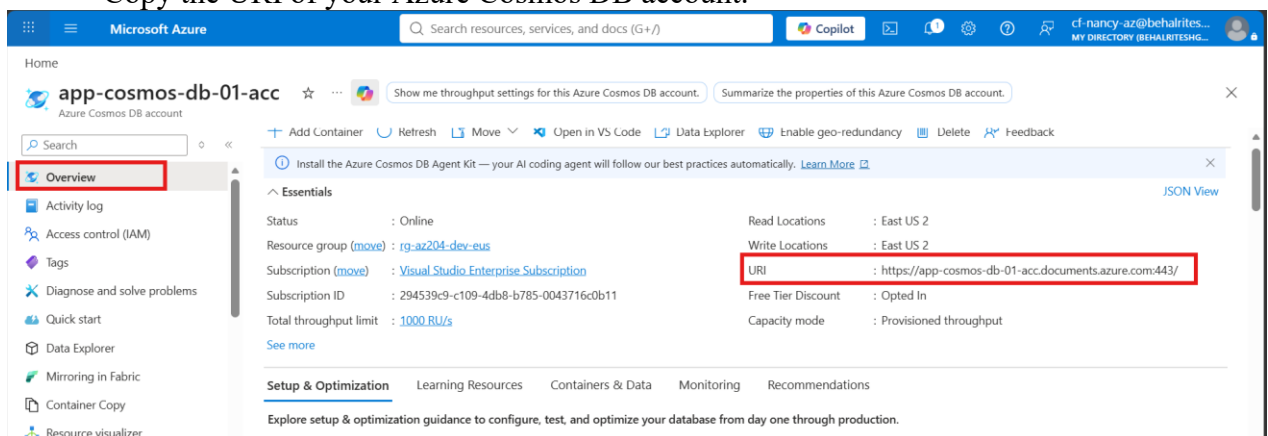
- Rename the function as **changeFeed**.



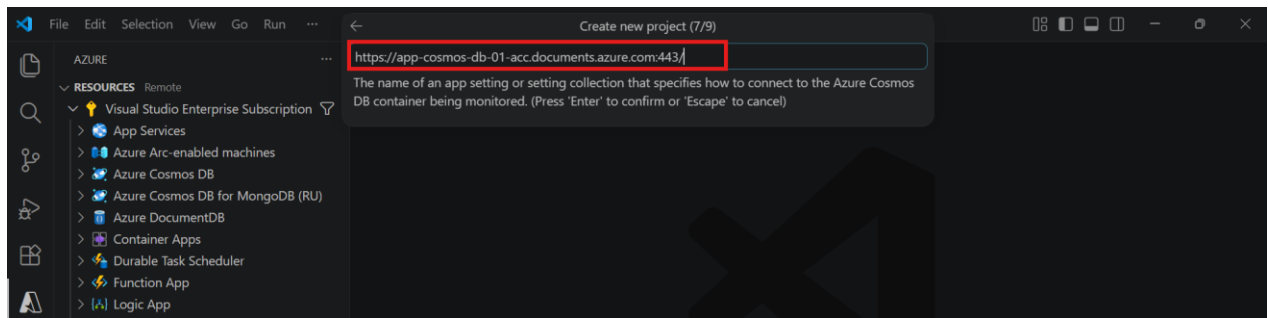
- Keep the namespace as default.



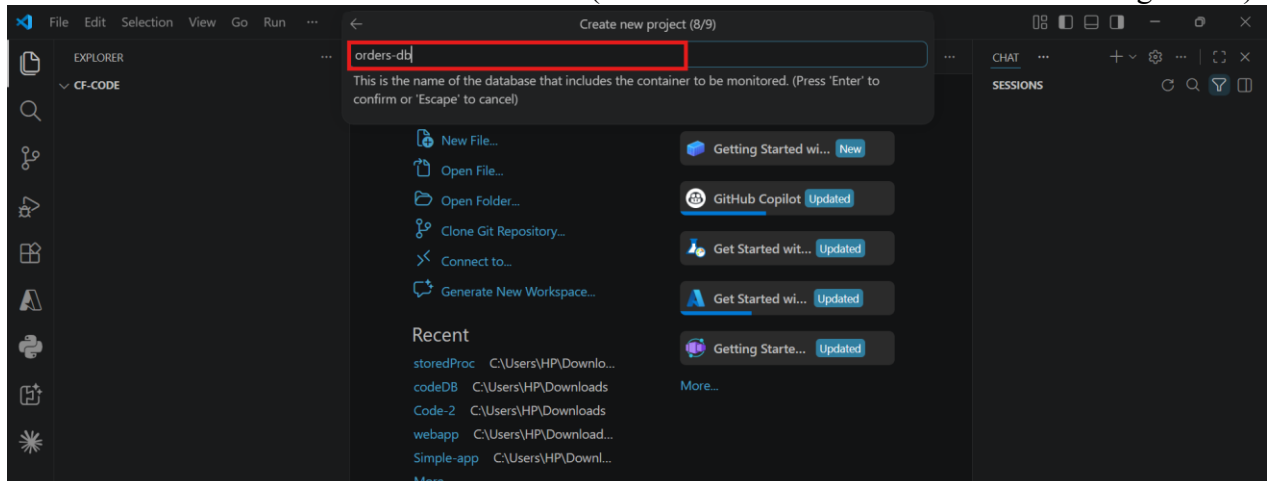
- Sign in as your Azure Account -if asked.
- Copy the URI of your Azure Cosmos DB account.



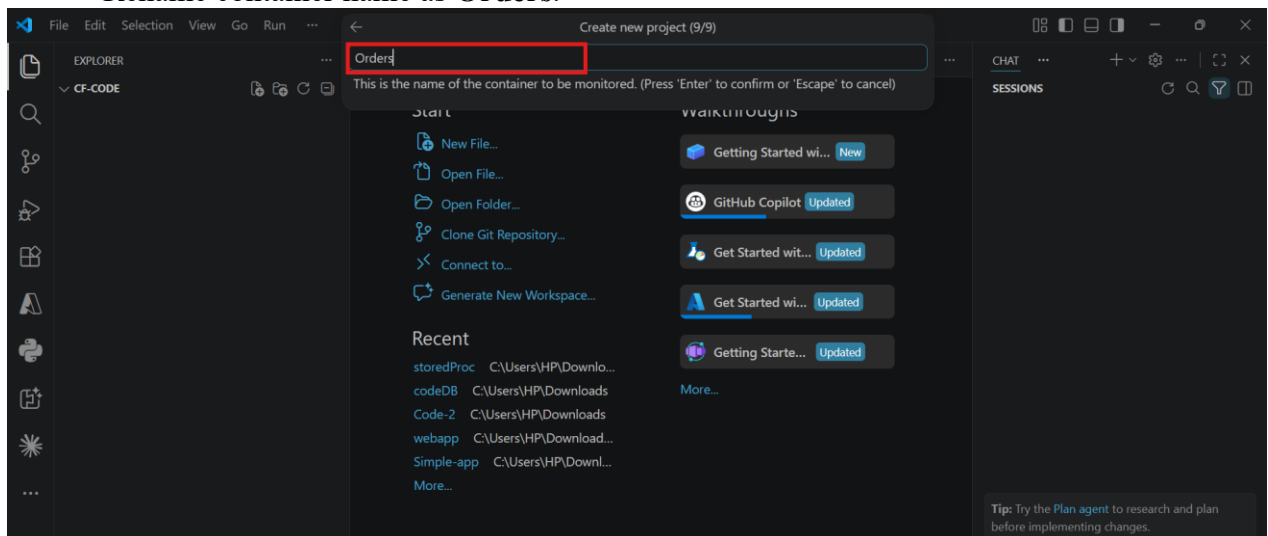
- Enter the URI in the Connection Account.



- Rename database name as **orders-db** (based on Azure cosmos DB account configuration).



- Rename container name as **Orders**.



- Press **ENTER**.
- The Azure Cosmos DB Trigger monitors the Change Feed and automatically executes the function whenever documents are created or updated in the configured container.

Step 3 – Configure the Azure Function

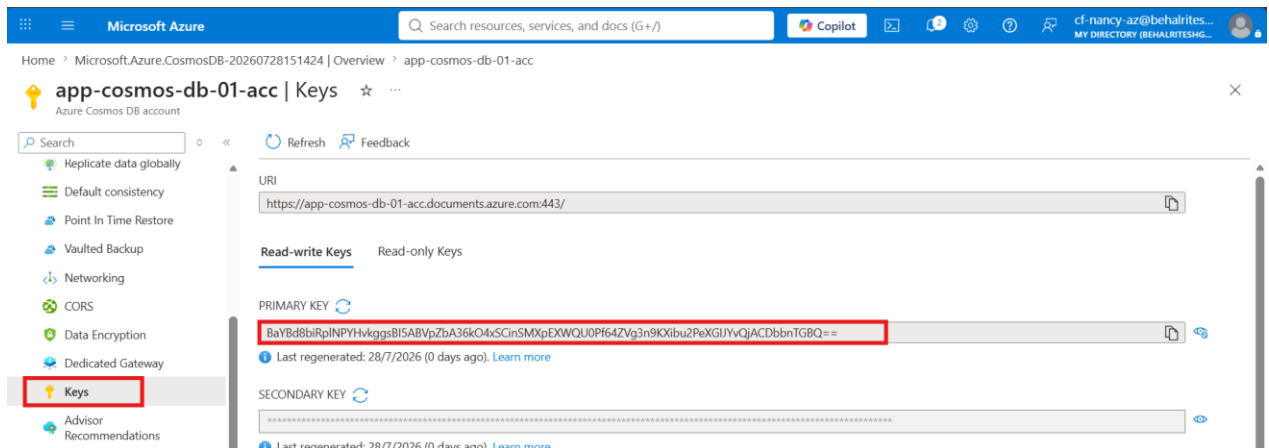
- **Open** the generated function file.
- **Replace** the existing code with the provided Azure Function code.

```
6 namespace Company.Function;
8 public class changeFeed
18 public void Run([CosmosDBTrigger(
21     Connection = "https://app-cosmos-db-01-acc.documents.azure.com:443/",
22     LeaseContainerName = "leases",
23     CreateLeaseContainerIfNotExists = true)] IReadOnlyList<Order> input)
24 {
25     if (input != null && input.Count > 0)
26     {
27         _logger.LogInformation("Documents modified: " + input.Count);
28         _logger.LogInformation("First document Id: " + input[0].id);
29     }
30 }
31 }
32
33 public class Order
34 {
35     public string id { get; set; } = default!;
36
37     public string customerId { get; set; } = default!;
38
39     public int type { get; set; } = default!;
40
41     public bool total { get; set; } = default!;
42 }
```

- Replace the **Connection** property (around line 21) with:
 - Connection = "CosmosDBConnection"

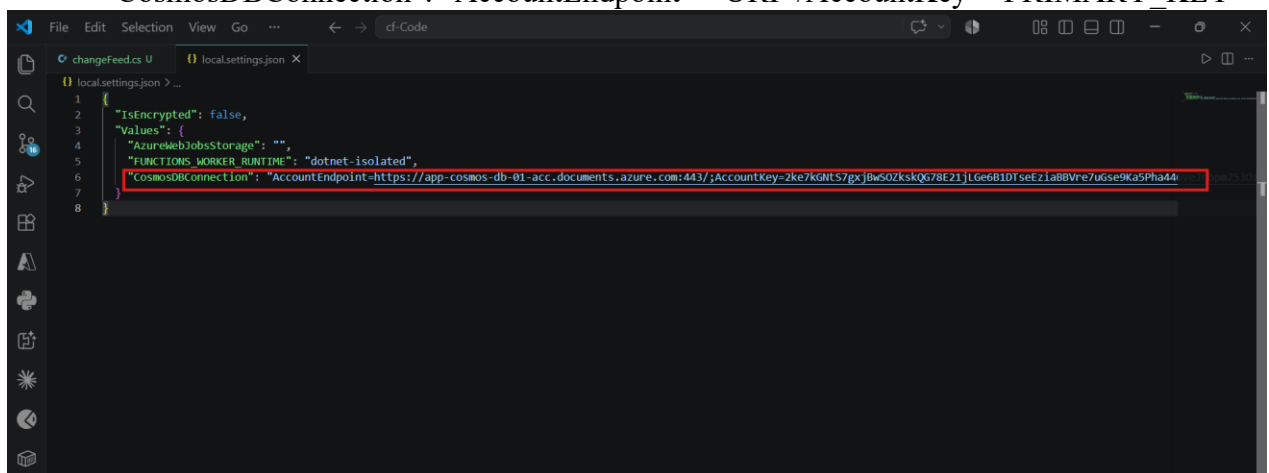
```
10 private readonly ILogger<changeFeed> _logger;
11
12 public changeFeed(ILogger<changeFeed> logger)
13 {
14     _logger = logger;
15 }
16
17 [Function("changeFeed")]
18 public void Run([CosmosDBTrigger(
19     databaseName: "orders-db",
20     containerName: "Orders",
21     Connection = "CosmosDBConnection",
22     LeaseContainerName = "leases",
23     CreateLeaseContainerIfNotExists = true)] IReadOnlyList<Order> input)
24 {
25     if (input != null && input.Count > 0)
26     {
27         _logger.LogInformation("Documents modified: " + input.Count);
28         _logger.LogInformation("First document Id: " + input[0].id);
29     }
30 }
31 }
```

- Open **local.settings.json**.
- Copy the Primary key of your Azure Cosmos DB account.



- **Enter:**

"CosmosDBConnection": "AccountEndpoint=<URI>/AccountKey=<PRIMARY_KEY>"



- **Save** all project files.
- The function contains the logic that processes documents received from the Azure Cosmos DB Change Feed.

Step 4 – Build and Run the Function

- Open the integrated terminal.
- Build the project:
 - dotnet build
 - dotnet run

```
File Edit Selection View Go Run ...
cf-Code
EXPLORER
CF-CODE
  _azure_db_table_json
  .gitignore
  AzuriteConfig
  cf-Code.csproj
  changeFeed.cs
  host.json
  local.settings.json
  Program.cs
OUTLINE
TIMELINE
CF PROJECT DETAILS
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE POSTMAN CONSOLE
PS C:\Users\HP\Downloads\GIT2\cosmos2\cf-Code> dotnet build
Restore complete (1.1s)
WorkerExtensions net8.0 succeeded (5.1s) -> obj\Debug\net10.0\WorkerExtensions\bin\Release\net8.0\Microsoft.Azure.Functions.Worker.Extensions.dll
cf-Code net10.0 succeeded (1.8s) -> bin\Debug\net10.0\cf-Code.dll

Build succeeded in 10.7s
PS C:\Users\HP\Downloads\GIT2\cosmos2\cf-Code> dotnet run
Using launch settings from C:\Users\HP\Downloads\GIT2\cosmos2\cf-Code\Properties\launchSettings.json...

Azure Functions Core Tools
Core Tools Version: 4.12.1+7f573ec7a43edc895383eb81b78fe86ead2b707b (64-bit)
Function Runtime Version: 4.1048.200.26180

master* 0 0 Projects: cf-Code Debug Any CPU C# Workspace Dev Mode Ln 32, Col 1 Spaces: 4 UTF-8 CRLF {} C# Go Live
File Edit Selection View Go Run ...
cf-Code
EXPLORER
CF-CODE
  _azure_db_table_json
  .gitignore
  AzuriteConfig
  cf-Code.csproj
  changeFeed.cs
  host.json
  local.settings.json
  Program.cs
OUTLINE
TIMELINE
CF PROJECT DETAILS
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE POSTMAN CONSOLE
PS C:\Users\HP\Downloads\GIT2\cosmos2\cf-Code> dotnet run

[2026-07-27T15:03:50.418Z] Found C:\Users\HP\Downloads\GIT2\cosmos2\cf-Code\cf-Code.csproj. Using for user secrets file configuration.
[2026-07-27T15:03:53.753Z] Worker process started and initialized.

Functions:

changeFeed: cosmosDBTrigger

For detailed output, run func with --verbose flag.
[2026-07-27T15:03:58.687Z] Host lock lease acquired by instance ID '000000000000000000000000000000003CF49764'.
[2026-07-27T15:04:08.446Z] [Tag=''] Process reporting unhealthy: Unhealthy. Health check entries are {"azure.functions.web_host.lifecycle":{"status":"Healthy","description":null},"azure.functions.script_host.lifecycle":{"status":"Healthy","description":null},"azure.functions.webjobs.storage":{"status":"Unhealthy","description":"Unable to create client for AzureWebJobsStorage"}}
Ln 32, Col 1 Spaces: 4 UTF-8 CRLF {} C# Go Live
```

- Allow the public and private network access to Azure func app (if pop-up occurs).
- The Azure Functions host starts listening for document changes in the configured Azure Cosmos DB container.

Step 5 – Test the Change Feed Trigger

- Add or update a document in the **Orders** container.

The image displays two screenshots of the Microsoft Azure Data Explorer interface, showing the 'app-cosmos-db-01-acc' account.

Top Screenshot:

- The 'Data Explorer' sidebar is on the left, with 'Items' selected under the 'Orders' database.
- The 'Home' tab is active, showing a query: `SELECT * FROM c`.
- The 'Orders.Items' tab is open, displaying a JSON document for an order with `"id": "ord-001"`.
- The 'Save' button is highlighted in the top toolbar.

Bottom Screenshot:

- The 'Data Explorer' sidebar is on the left, with 'Items' selected under the 'Orders' database.
- The 'Home' tab is active, showing a query: `SELECT * FROM c`.
- The 'Orders.Items' tab is open, displaying a JSON document for an order with `"id": "ord-002"`.
- The 'New Item' button is highlighted in the top toolbar.

- Observe the Azure Functions terminal.

The screenshot shows the Visual Studio Code interface with the 'cf-Code' project open. The Explorer pane on the left shows the project structure, including 'changeFeed.cs' and 'local.settings.json'. The main editor displays the code for 'changeFeed.cs', which defines a 'Company.Function' namespace and a 'changeFeed' class. The class has a 'Run' method that is triggered by a Cosmos DB trigger named 'orders-db'. The terminal pane at the bottom shows the output of the 'dotnet run' command, indicating that the function was executed successfully. A red box highlights the log entry: '[2026-07-27T15:37:10.473Z] Documents modified: 1'.

- **Verify that the function executes automatically.**

This screenshot is similar to the one above, showing the Visual Studio Code interface with the 'cf-Code' project. The terminal pane shows the output of the 'dotnet run' command, indicating that the function was executed successfully. A red box highlights the log entry: '[2026-07-27T15:42:37.125Z] First document Id: ord-002'.

- Whenever a document is inserted or modified, Azure Cosmos DB records the change in the Change Feed, and the Azure Cosmos DB Trigger automatically invokes the function without requiring manual polling.

Verification

Verify the following:

- The Change Feed feature is enabled.
- The Azure Functions project is created successfully.
- The Azure Cosmos DB Trigger is configured.
- The function builds and runs successfully.

- The function executes automatically after documents are created or updated in the **Orders** container.
- The terminal displays the processed Change Feed events successfully.